

Transaction Cost Analysis

2024-04-01 to 2024-06-14 · Demo Client

Prepared for	Demo Client
Engagement	DEMO-0008
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Executive summary

Demonstration report. Built entirely from synthetic data generated by ProQuant for illustration. The manager, the brokers, the stocks, the orders and the prices are fictitious.

ProQuant Analytics measured the execution of 1,139 orders of Demo Client (2024-04-01 to 2024-06-14): every order is compared with the price when the decision was taken, the price when the order reached the desk, the market VWAP while it was traded and the price afterwards; its cost is split into delay, execution, opportunity and explicit costs; the costs are explained by a pre-trade model so that brokers, algorithms and desks are compared fairly; the best-execution policy is applied order by order; and the savings within reach are estimated. Costs are in basis points (bps) of the paper notional; positive numbers are costs.

Overall view: Execution quality raises material concerns

7 of 22 tests passed, 13 with a warning, 2 failed. Conclusions are drafted from the figures and reviewed by the analyst before release.

ORDERS MEASURED
1,139
4.11bn traded · 55 days

IMPLEMENTATION SHORTFALL
22.4 bps
median 14.5 · 95th pct 78

SLIPPAGE VS ARRIVAL
17.9 bps
vs VWAP +17.8 · 8% beat it

EXPLICIT COSTS
1.61 bps
commissions and fees

WORST BROKER (MODEL-ADJUSTED)
+17.2 bps
Gamma Capital

SAVINGS WITHIN REACH
6.1 bps
12.3m a year

Conclusions

- **Headline** 1,139 orders (1,171 received, 32 excluded) with a traded notional of USD 4.11bn between 2024-04-01 and 2024-06-14 cost 22.4 bps of implementation shortfall (weighted by notional): delay +1.7, execution +16.6, opportunity +2.5 and explicit costs 1.6 bps.
- **Benchmarks** Against the arrival price the slippage is 17.9 bps and against the interval VWAP +17.8 bps; 8% of the orders beat the VWAP. After the last fill the price moves back by 17.2 bps, 96% of the slippage: the cost is mostly temporary impact, not information.
- **Brokers** Against what the model expects for the difficulty of the orders, Gamma Capital is +17.2 bps (worse than the model; interval +9.3 to +25.6) and Beta Markets -5.6 bps (not significant). Raw shortfall: Beta Markets 17.5, Alpha Securities 19.0, Delta Direct 21.0, Gamma Capital 34.9 bps.
- **Algorithms** MARKET is the most expensive algorithm against the model (+14.2 bps, worse than the model) and VWAP the cheapest (-6.0 bps, not significant).
- **Size** Cost rises with the square root of the order size and the volatility: the impact coefficient is 0.50 (t = 6.2); the model explains 1.5% of the variation out of sample, which is normal because the market moves during the execution. Orders above 5% of the daily volume cost +24.5 bps more than those below 0.5%.
- **Timing** 7% of the notional trades in the opening half hour. The slowest desk (PM-Macro) waits 13 minutes on average between the decision and the arrival and pays +4.1 bps of delay cost, against +0.2 bps for PM-Growth.

- **Policy** Best-execution policy: 25 order(s) filled beyond their limit price (mostly Gamma Capital), 171 above the participation cap, 83 above their slippage limit, 68 with a delay above the limit and 235 above the commission cap. The largest broker has 39% of the notional.
- **Data** 32 order(s) were excluded because they cannot be measured honestly: fill before the order arrived (clock) (12); no market data (10); fill price outside the bar range (6); filled for more than the order quantity (4). 10 duplicate fill(s) were removed and 15 order(s) use the arrival price because the decision price is missing.
- **Savings** Savings within reach are estimated at 6.1 bps of notional, about USD 12.3m a year (upper bound 8.7 bps, as the scenarios overlap); the largest are 'Brokers to the median' (4.1 bps) and 'Algorithms to the median' (1.7 bps).
- **Overall** Execution quality raises material concerns: 7 tests passed, 13 with a warning, 2 failed. Priority actions: Share the broker scorecard; agree a remediation plan or reduce its allocation. Escalate to compliance: require the broker to respect limit prices and report each breach.
- **Analyst** Illustrative note (synthetic data): the head of trading confirms that the DMA flow of Delta Direct is used for urgent orders only, which explains part of its cost against the model.
- **Analyst** Illustrative note (synthetic data): the compliance officer has been told about the limit-price fills of Gamma Capital and has asked the broker for an explanation in writing.

Actions

Test	Status	Action
Worst broker against the model	FAIL	Share the broker scorecard; agree a remediation plan or reduce its allocation.
Orders filled beyond their limit	FAIL	Escalate to compliance: require the broker to respect limit prices and report each breach.
Orders with market data	WARN	Complete the market data (halted or missing symbols) or drop those orders from the policy scope.
Orders with a decision price	WARN	Capture the decision price in the OMS at the moment the PM decides.
Implementation shortfall (weighted)	WARN	Review the largest contributors (broker, algorithm, size); see the savings table.
Slippage against the arrival price	WARN	Review algorithms and brokers with the highest alpha against the model.
Cost against the interval VWAP	WARN	Check that VWAP orders are not front-loaded and that urgent orders are not using passive algorithms.
Notional with a shortfall above 75 bps	WARN	Review the worst orders (order review sheet): size, urgency, algorithm.
Opportunity cost (unfilled)	WARN	Review limit levels and the handling of unfilled orders.
Gap between algorithms against the model	WARN	Set rules for which algorithm to use by urgency and size.
Orders above the participation cap	WARN	Slow the order down or split it over several days.
Orders above their slippage limit	WARN	Document the reason for each order above its limit in the order review sheet.
Share of the cost that reverts afterwards	WARN	Most of the cost is temporary impact: slow down or use passive strategies for large orders.
Extra cost of the largest orders	WARN	Cap the size of an order as a share of the average daily volume.
Savings within reach	WARN	Prioritise the scenarios with the largest saving in the savings table.

Source: tables/tests.csv

Methodology

This report tells what was found. The steps, formulas and parameters are in the methodology document. One thing helps when reading on: alpha is an order's actual cost minus what the pre-trade model expects for an order of that difficulty, so a group's alpha can be compared fairly with another's.

Results

1. Overall cost

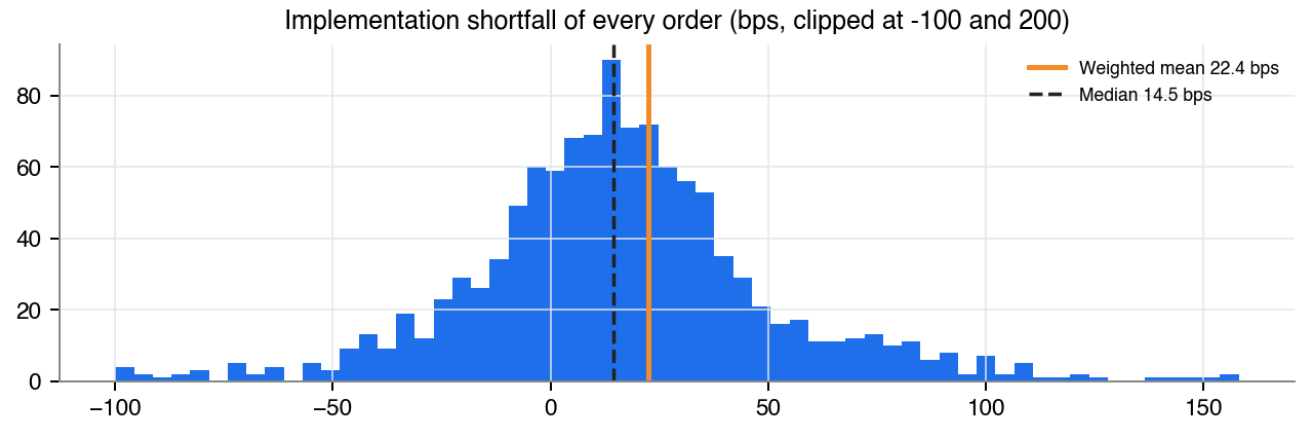


Figure 1. Shortfall of every order. Source: tables/orders_costs.csv

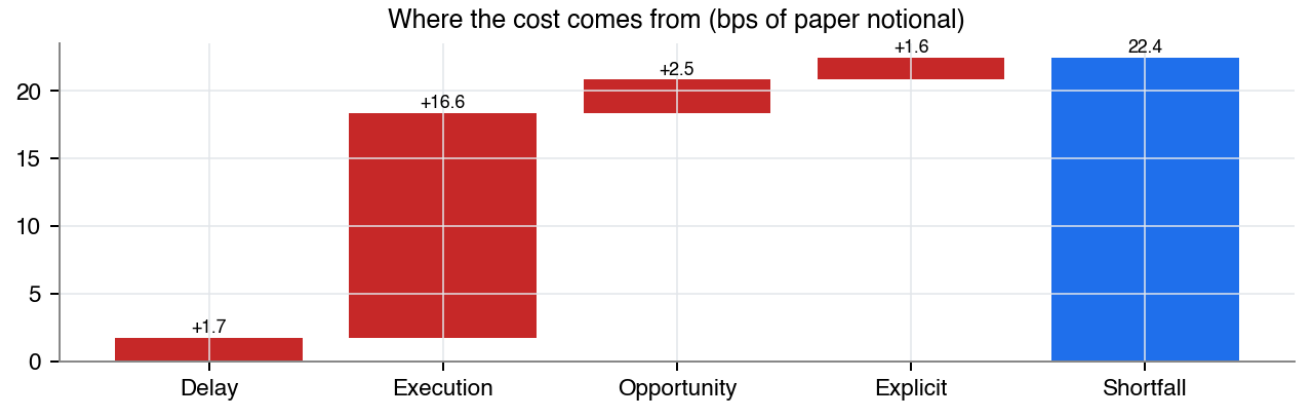


Figure 2. Delay, execution, opportunity and explicit costs. Source: tables/orders_costs.csv

Weighted by notional: shortfall **22.4 bps** (median 14.5), of which delay +1.7, execution +16.6, opportunity +2.5 and explicit 1.6. Slippage against the arrival price 17.9 bps; against the interval VWAP +17.8 bps.

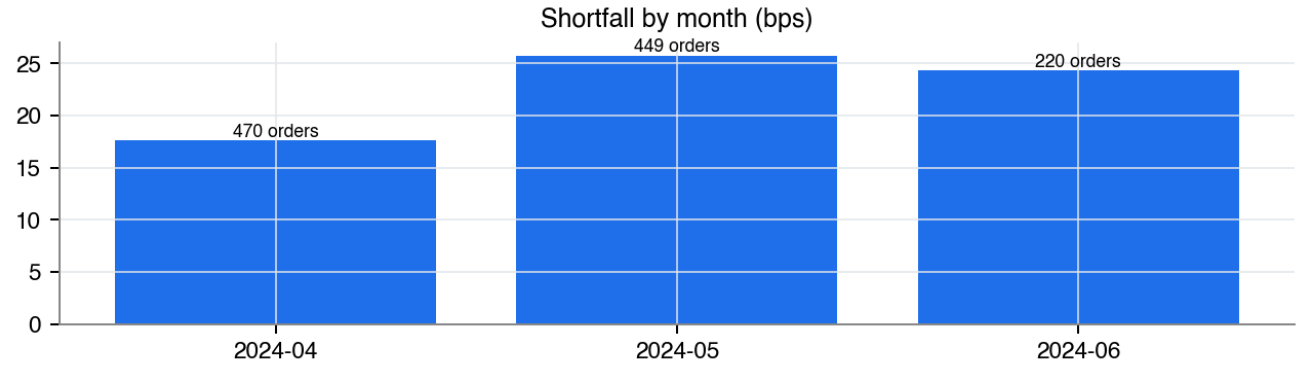


Figure 3. Shortfall by month. Source: tables/by_month.csv

2. Brokers

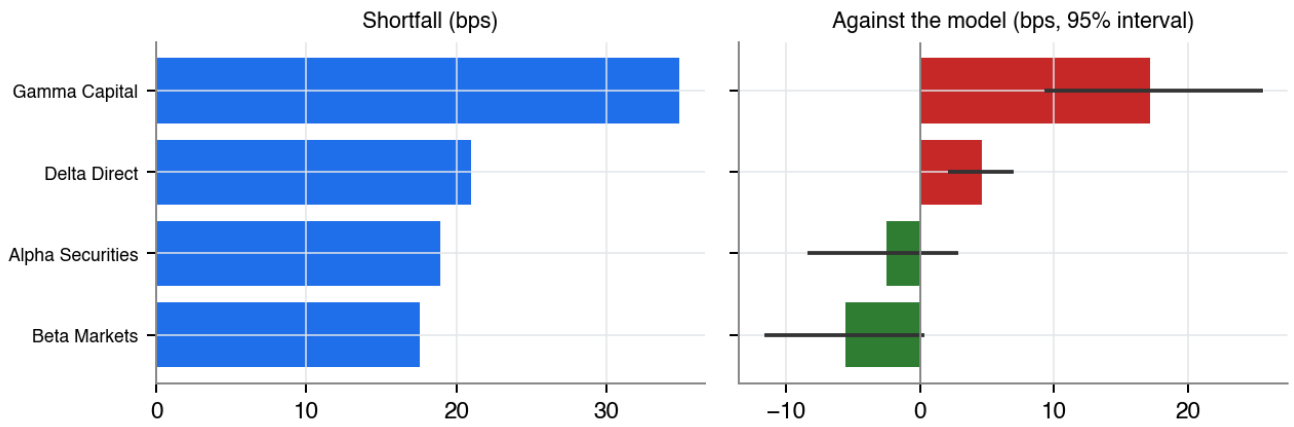


Figure 4. Raw shortfall and shortfall against the model. Source: tables/by_broker.csv

Broker	Orders	Notional	Shortfall	vs arrival	vs VWAP	Explicit	Reversion	Model	Alpha	Low	High	Result
Gamma Capital	235	22%	34.9	33.2	+25.0	2.5	15.7	16.0	+17.2	9.3	25.6	worse than the model
Delta Direct	159	13%	21.0	19.5	+20.6	0.8	18.8	14.9	+4.6	2.1	7.0	worse than the model
Alpha Securities	418	39%	19.0	13.7	+14.7	1.6	19.2	16.2	-2.5	-8.4	2.8	not significant
Beta Markets	327	26%	17.5	10.3	+14.8	1.3	15.0	15.8	-5.6	-11.6	0.4	not significant

Source: tables/by_broker.csv

Model = what the model expects for the difficulty of the orders (size, volatility, spread, urgency). Alpha = actual slippage minus the model: positive is worse. "Worse than the model" means the 95% bootstrap interval is above zero.

Broker and algorithm

Broker	Algorithm	Orders	Notional	Shortfall	vs arrival	vs VWAP	Explicit	Reversion	Model	Alpha	Low	High	Result
Gamma Capital	POV	76	7%	46.7	43.3	+26.5	2.5	18.7	16.2	+27.1	8.0	42.9	worse than the model
Gamma Capital	MARKET	93	9%	29.4	30.8	+28.7	2.7	22.3	16.6	+14.2	10.7	17.8	worse than the model
Gamma Capital	TWAP	66	6%	28.3	23.8	+17.0	2.4	0.5	14.7	+9.1	-4.2	21.6	not significant
Delta Direct	DMA	159	13%	21.0	19.5	+20.6	0.8	18.8	14.9	+4.6	2.1	7.0	worse than the model
Alpha Securities	IS	232	18%	19.7	15.2	+13.1	1.6	12.4	14.5	+0.8	-5.0	5.6	not significant
Beta Markets	TWAP	92	8%	17.7	15.0	+14.3	1.2	8.3	17.3	-2.3	-16.8	9.4	not significant
Alpha Securities	VWAP	186	20%	18.3	12.4	+16.1	1.6	25.4	17.9	-5.5	-15.7	2.6	not significant
Beta Markets	POV	142	11%	14.0	8.2	+15.9	1.3	16.7	14.9	-6.6	-16.9	3.1	not significant
Beta Markets	VWAP	93	7%	22.7	8.4	+13.5	1.2	19.2	15.9	-7.4	-16.9	1.9	not significant

Source: tables/by_broker_algo.csv

3. Algorithms

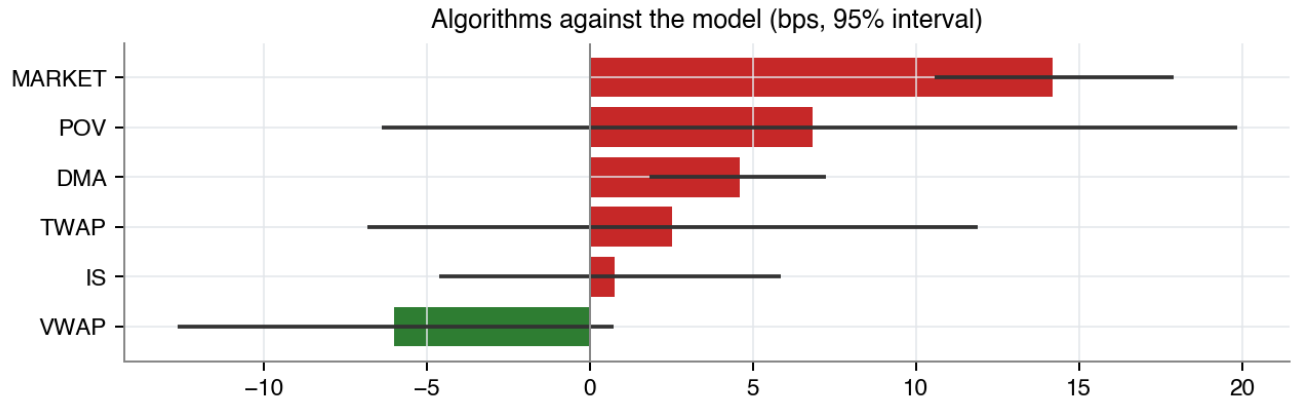


Figure 5. Algorithms against the model. Source: tables/by_algo.csv

Algorithm	Orders	Notional	Shortfall	vs arrival	vs VWAP	Explicit	Reversion	Model	Alpha	Low	High	Result
MARKET	93	9%	29.4	30.8	+28.7	2.7	22.3	16.6	+14.2	10.6	17.9	worse than the model
POV	218	19%	27.0	22.2	+20.1	1.8	17.5	15.4	+6.8	-6.4	19.8	not significant
DMA	159	13%	21.0	19.5	+20.6	0.8	18.8	14.9	+4.6	1.8	7.2	worse than the model
TWAP	158	13%	22.2	18.7	+15.5	1.7	5.0	16.2	+2.5	-6.8	11.9	not significant
IS	232	18%	19.7	15.2	+13.1	1.6	12.4	14.5	+0.8	-4.6	5.9	not significant
VWAP	279	28%	19.5	11.3	+15.4	1.5	23.7	17.3	-6.0	-12.6	0.7	not significant

Source: tables/by_algo.csv

4. Size, volatility and the model

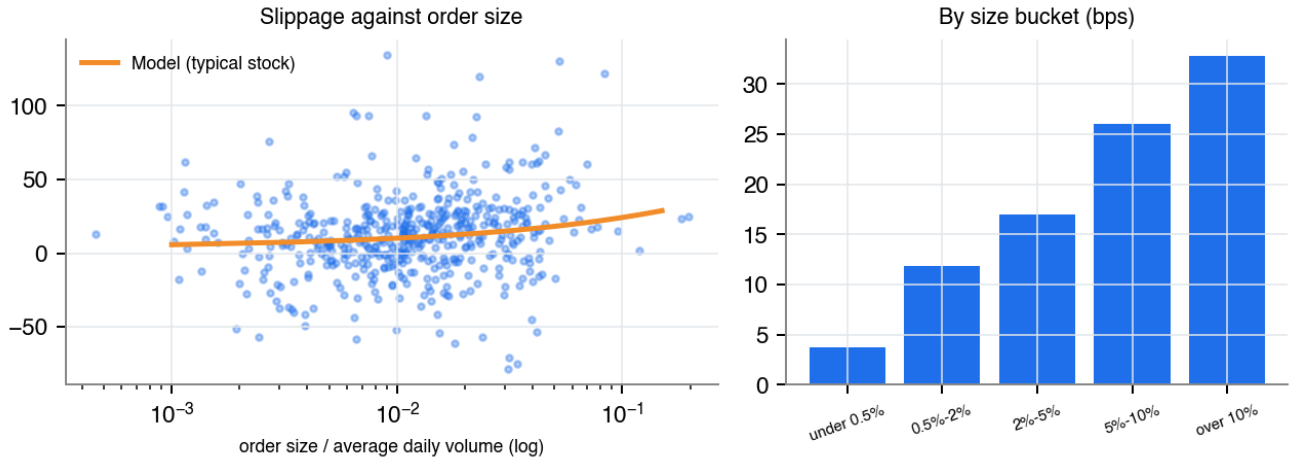


Figure 6. Slippage against order size and the fitted model. Source: tables/orders_costs.csv, tables/model_curve.csv

Size (% of ADV)	Orders	Notional	Shortfall	vs arrival	vs VWAP	Explicit	Reversion	Model	Alpha	Low	High	Result
over 10%	14	9%	33.1	32.8	+34.3	1.9	46.5	25.0	+7.9	-24.9	35.3	not significant
5% to 10%	58	18%	31.9	26.0	+24.3	1.7	22.2	21.7	+4.3	-6.5	13.1	not significant
2% to 5%	257	38%	22.5	16.9	+16.8	1.6	15.4	15.7	+1.2	-3.5	6.1	not significant
0.5% to 2%	588	32%	15.7	11.8	+11.6	1.6	9.1	11.2	+0.6	-1.7	2.7	not significant
under 0.5%	222	3%	6.5	3.7	+8.6	1.6	11.2	7.7	-4.0	-8.4	-0.1	better than the model

Source: tables/by_size.csv

The model

Term	Coefficient	Std error	t	Training sample
intercept	2.937	2.454	1.2	2.619
impact_term	0.498	0.081	6.2	0.524
half_spread	0.340	0.672	0.5	0.024
urgent	-1.012	2.377	-0.4	-1.581
patient	0.753	1.842	0.4	-0.016

Source: tables/model_coefficients.csv

Out-of-sample R^2 is 1.5% (training 3.5%, 773 orders to 2024-05-22; 366 orders after). A low R^2 is normal in this work: the market moves by far more than the impact of one order while it is executed. What counts is that the size coefficient is positive and significant (t = 6.2) and that the alpha of the groups is measured with a confidence interval.

5. Time of day, desks, urgency and side

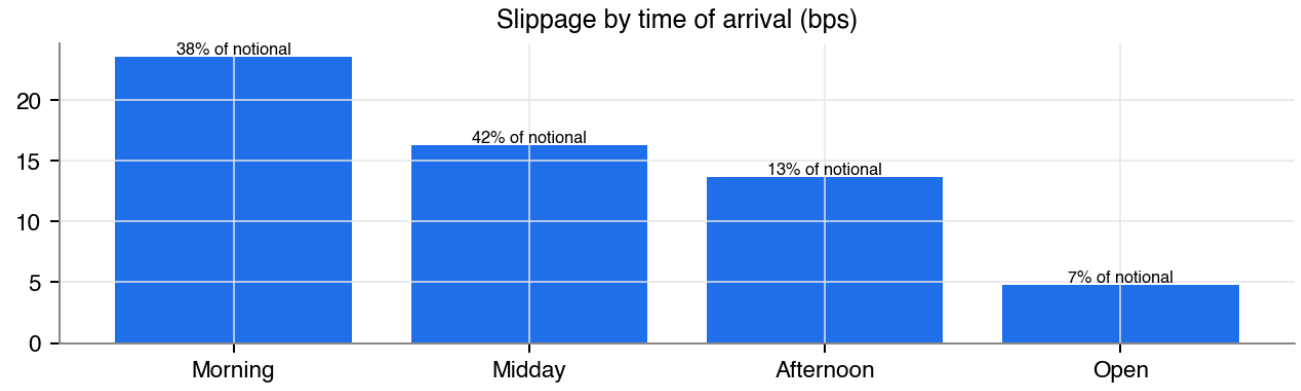


Figure 7. Slippage by arrival time. Source: tables/by_time_of_day.csv

Arrival time	Orders	Notional	Shortfall	vs arrival	vs VWAP	Explicit	Reversion	Model	Alpha	Low	High	Result
Morning (10:00-12:00)	440	38%	27.4	23.6	+19.5	1.7	19.3	15.1	+8.4	2.5	14.5	worse than the model
Midday (12:00-14:00)	455	42%	21.8	16.3	+16.8	1.6	16.5	17.0	-0.7	-6.6	5.4	not significant
Afternoon (14:00-15:30)	159	13%	16.5	13.7	+17.1	1.6	10.2	14.8	-1.1	-7.9	6.3	not significant
Open (09:30-10:00)	85	7%	9.8	4.7	+15.6	1.5	24.0	15.9	-11.2	-21.4	1.2	not significant

Source: tables/by_time_of_day.csv

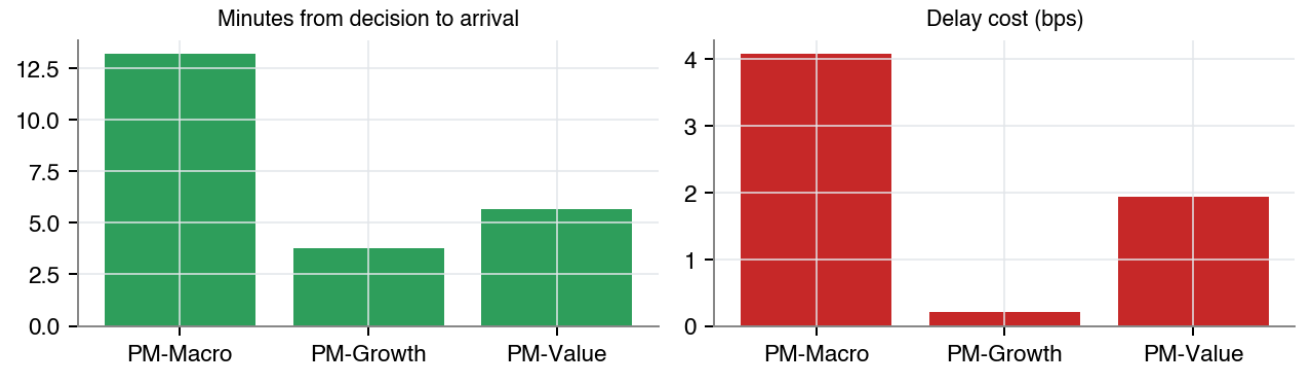


Figure 8. Time to arrival and delay cost by desk. Source: tables/by_portfolio.csv

Desk	Orders	Notional	Shortfall	vs arrival	vs VWAP	Explicit	Reversion	Model	Alpha	Low	High	Result
PM-Macro	278	25%	30.6	22.2	+18.2	1.7	14.2	16.7	+5.5	-2.5	14.8	not significant
PM-Growth	452	42%	21.1	18.3	+18.6	1.6	19.3	16.4	+1.9	-2.6	6.1	not significant
PM-Value	409	34%	18.1	14.3	+16.4	1.6	16.9	14.7	-0.4	-6.4	5.5	not significant

Source: tables/by_portfolio.csv

Urgency	Orders	Notional	Shortfall	vs arrival	vs VWAP	Explicit	Reversion	Model	Alpha	Low	High	Result
high	167	13%	24.0	20.6	+19.0	1.6	22.4	14.4	+6.2	0.9	10.7	worse than the model
normal	633	55%	21.8	18.2	+16.7	1.6	14.6	14.9	+3.2	-2.2	8.5	not significant
low	339	31%	22.8	16.3	+19.1	1.6	19.8	18.3	-2.0	-8.8	4.8	not significant

Source: tables/by_urgency.csv

Side	Orders	Notional	Shortfall	vs arrival	vs VWAP	Explicit	Reversion	Model	Alpha	Low	High	Result
Buy	556	49%	26.0	20.7	+19.0	1.7	15.4	16.4	+4.3	-1.7	10.3	not significant
Sell	583	51%	19.0	15.2	+16.6	1.6	19.1	15.5	-0.3	-4.0	3.6	not significant

Source: tables/by_side.csv

6. Venues and reversion

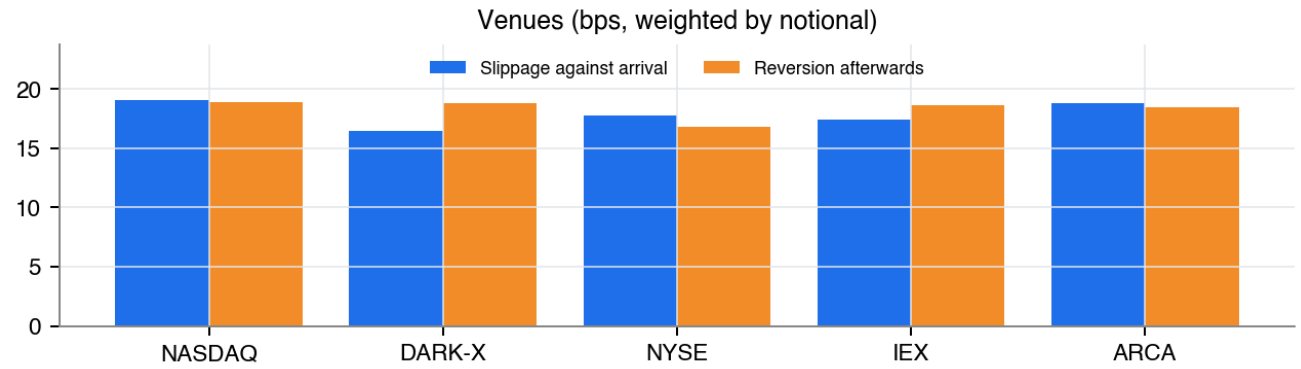


Figure 9. Venues. Source: tables/by_venue.csv

Venue	Fills	Notional	Slippage	Reversion
NASDAQ	3,129	24%	19.0	18.9
DARK-X	3,050	24%	16.4	18.8
NYSE	2,538	21%	17.7	16.8
IEX	1,915	16%	17.5	18.6
ARCA	1,831	15%	18.8	18.5

Source: tables/by_venue.csv

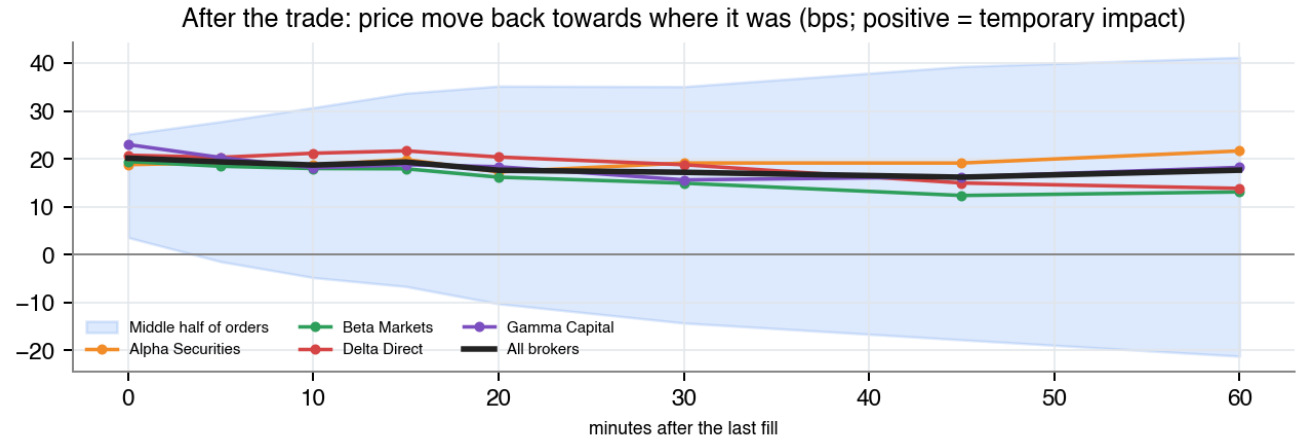


Figure 10. Price after the trade. Source: tables/reversion_profile.csv

If the price moves back after the last fill, part of the cost was temporary impact: a slower or more patient execution would have cost less. A move that does not come back means the order carried information.

7. Best-execution policy

Limits: slippage 60 / 50 / 40 bps (low / normal / high urgency), participation 50%, delay 20 minutes, commissions 2.0 bps, no fill beyond the limit price, no broker above 40% of the notional.

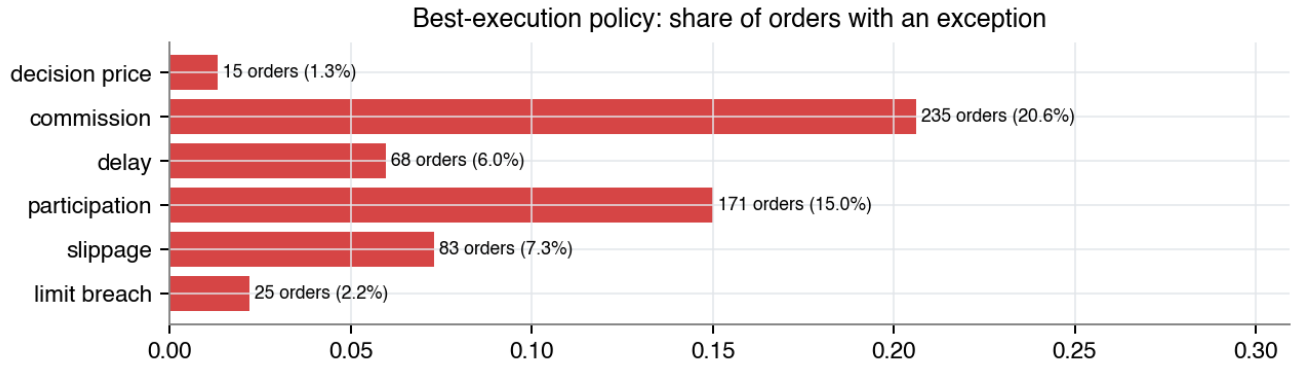


Figure 11. Orders with an exception, by rule. Source: tables/policy_summary.csv

Rule	Orders	Share of orders	Share of notional	Most exceptions
limit_breach	25	2.2%	2.1%	Gamma Capital
slippage	83	7.3%	14.0%	Gamma Capital
participation	171	15.0%	28.4%	Delta Direct
delay	68	6.0%	8.4%	Alpha Securities
commission	235	20.6%	22.6%	Gamma Capital
decision_price	15	1.3%	1.2%	Alpha Securities

Source: tables/policy_summary.csv

Each exception, with its order, is in the best-execution report and the order review sheet.

8. Savings within reach

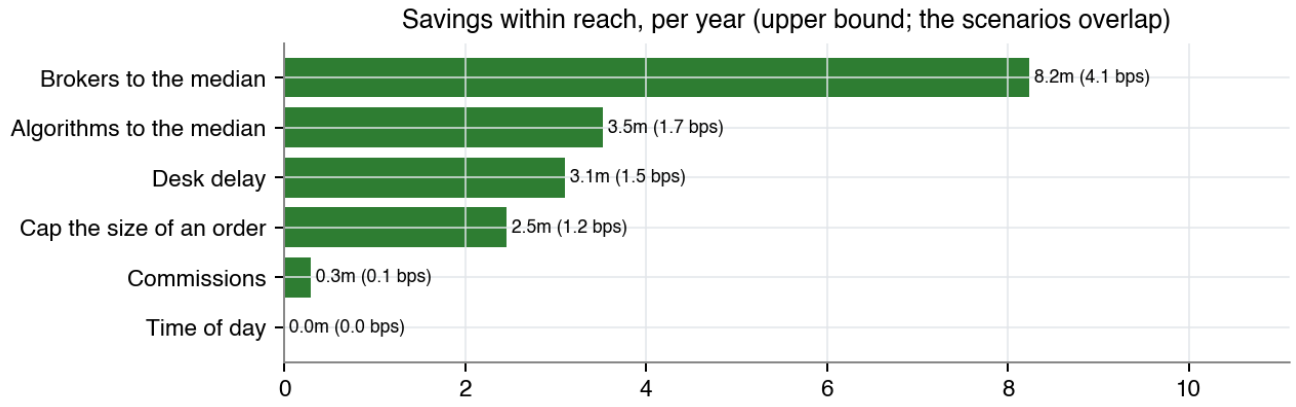


Figure 12. Savings by scenario. Source: tables/savings.csv

Scenario	What if	bps	Per year	Confidence
Brokers to the median	Orders of Gamma Capital, Delta Direct cost as much as the median broker (alpha +1.0 bps)	4.1	8,234,427	medium
Algorithms to the median	Orders of MARKET, POV cost as much as the median algo (alpha +3.6 bps)	1.7	3,518,415	low
Cap the size of an order	Orders above 5% of the average daily volume (72 orders) are split so that none is above it	1.2	2,459,873	low
Desk delay	Desks slower than the fastest one (+0.2 bps of delay cost) are brought to it	1.5	3,099,497	medium
Commissions	Commissions above 2.0 bps are brought to the cap	0.1	298,093	medium
Time of day	50% of the opening half hour moves to midday (-11.6 bps difference in slippage)	0.0	0	low

Source: tables/savings.csv

Total: 8.7 bps if every scenario worked independently (upper bound); 6.1 bps after allowing for overlap (a factor of 0.7). These are estimates for the period, scaled to 252 trading days, not forecasts.

9. Data quality

1,171 orders and 12,737 fills were received; 32 orders were excluded, 10 duplicate fills removed and 15 orders use the arrival price in place of a missing decision price.

Level	Reason	Count
fill	fill before the order arrived	12
fill	no market data at the time of the fill	108
fill	price outside the bar range	6
order	fill before the order arrived (clock)	12
order	fill price outside the bar range	6
order	filled for more than the order quantity	4
order	no market data	10

Source: tables/data_exceptions.csv

10. All tests

Area	Test	Result	Threshold	Status
Data	Orders that can be measured	97.3%	warn below 97%, fail below 93%	PASS
Data	Fills with a valid price and time	0.14%	warn above 0.20%, fail above 1.00%	PASS
Data	Orders with market data	0.85%	warn above 0.00%, fail above 1.00%	WARN
Data	Orders with a decision price	1.32%	warn above 1.00%, fail above 3.00%	WARN
Cost	Implementation shortfall (weighted)	22.4 bps	warn above 15.0 bps, fail above 30.0 bps	WARN
Cost	Slippage against the arrival price	17.9 bps	warn above 10.0 bps, fail above 20.0 bps	WARN
Cost	Cost against the interval VWAP	17.8 bps	warn above 8.0 bps, fail above 20.0 bps	WARN
Cost	Notional with a shortfall above 75 bps	8.7%	warn above 5.0%, fail above 15.0%	WARN
Cost	Commissions and fees	1.61 bps	warn above 2.50 bps, fail above 4.00 bps	PASS
Cost	Delay cost	1.7 bps	warn above 2.0 bps, fail above 6.0 bps	PASS
Cost	Opportunity cost (unfilled)	2.5 bps	warn above 2.0 bps, fail above 8.0 bps	WARN
Brokers	Worst broker against the model	Gamma Capital: +17.2 bps (worse than the model; interval +9.3 to +25.6)	counts only if significant; warn above 3 bps, fail above 8	FAIL
Brokers	Largest broker's share of notional	38.6%	warn above 40.0%, fail above 60.0%	PASS
Brokers	Gap between algorithms against the model	20.2 bps	warn above 8.0 bps, fail above 35.0 bps	WARN
Policy	Orders filled beyond their limit	2.19%	warn above 0.00%, fail above 1.00%	FAIL
Policy	Orders above the participation cap	15.0%	warn above 2.0%, fail above 20.0%	WARN
Policy	Orders above their slippage limit	7.3%	warn above 5.0%, fail above 15.0%	WARN
Impact	Share of the cost that reverts afterwards	96%	warn above 60%, fail above 200%	WARN
Impact	Extra cost of the largest orders	24.5 bps	warn above 10.0 bps, fail above 25.0 bps	WARN
Impact	Size matters (impact coefficient)	coefficient 0.50, t = 6.2; R^2 out of sample 1.5%	positive with t above 2	PASS
Impact	Notional traded in the opening half hour	6.8%	warn above 25.0%, fail above 40.0%	PASS
Impact	Savings within reach	6.1 bps	warn above 3.0 bps, fail above 12.0 bps	WARN

Source: tables/tests.csv

Limitations and assumptions

- Prices and volumes come from 5-minute bars: a fill is compared with the open of its bar. Costs of a few basis points are within the resolution of the data.
- The decision price is what the order management system recorded; where it is missing the arrival price is used, which hides any delay cost.
- Slippage is noisy: the market moves during the execution. Averages are weighted by notional and intervals are bootstrapped; a difference inside its interval is not evidence.
- The pre-trade model explains a small share of the variation of single orders and is meant for comparing groups of orders, not for quoting the cost of one.
- The savings are scenario estimates that overlap; they are not a forecast and not additive.
- The policy limits are those agreed at kickoff. An exception is a reason to ask, not proof of a breach of duty.

Appendix

Cleaning log

step	action	rows_affected
harmonise	exchange suffix removed from the symbol of fills (e.g. .US)	190
duplicates	exact duplicate fills removed	10
harmonise	side read from B / S / Buy / Sell spellings	277
exclude	order excluded: fill before the order arrived (clock)	12
exclude	order excluded: no market data	10
exclude	order excluded: fill price outside the bar range	6
exclude	order excluded: filled for more than the order quantity	4
fallback	decision price missing: the arrival price will be used (order flagged)	15

Source: tables/cleaning_log.csv

Data quality

bars · 85566 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

fills · 12737 rows · PASSED (0 errors, 2 warnings)

Check	Severity	Column	Count	Message
duplicate_rows	warning		10	10 fully duplicated rows
duplicate_keys	warning		20	20 rows share a key ['fill_id'] (0 with conflicting values)

Source: data_quality/dq_fills.json

orders · 1171 rows · PASSED (0 errors, 0 warnings)

Check	Severity	Column	Count	Message
missing_values	info	decision_price	15	15 missing (1.28%)
missing_values	info	limit_price	783	783 missing (66.87%)

Source: data_quality/dq_orders.json

orders_clean · 1139 rows · PASSED (0 errors, 0 warnings)

Check	Severity	Column	Count	Message
missing_values	info	decision_price	15	15 missing (1.32%)
missing_values	info	limit_price	765	765 missing (67.16%)

Source: data_quality/dq_orders_clean.json

Audit trail

Run ID	tc_20261111T142137Z_e0c00a
Run started / finished (UTC)	2026-11-11T14:21:37+00:00
Analyst	info@proquantanalytics.com
Code version (git)	3a6bb15275c5 (uncommitted changes) · core 0.1.0
Config SHA-256	d3e18f911785c174
Input: orders	1171 rows · sha256 385479b70739727c
Input: fills	12737 rows · sha256 9f474f09f7fed8bf
Input: bars	85566 rows · sha256 944f1f24550488b9
Environment	python 3.13.5 · pandas 2.2.3 · numpy 2.1.3

Source: audit.json in the run folder

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